

## Project Proposal Summary

- **Problem Formulation** (most important): Clear business problem + what you aim to explore (patterns, trends, anomalies, relationships)

E-commerce is an ever-changing industry that continues to expand and attract new leads worldwide. Such growth encompasses a variety of customers who are continually evolving. For that reason, businesses face the unique challenge of understanding their customers' behavior as a crucial piece for better positioning themselves, allocating resources for operational excellence, and understanding seasonal patterns to maximize results.

This study aims to:

- Segment customers based on spending, recency, and frequency of purchase behaviors.
- Highlight operational efficiency in diverse product categories.
- Pinpoint sales trends for seasonal effects, Pareto distribution, and customer retention.

- **Dataset Overview: Source + key variables + why it fits the problem**

The dataset to be explored is the Online Retail II UCI, a real online retail transaction data set of two years from the digital repository of Kaggle. This dataset contains 1,048,576 rows and 8 columns. The columns are described as follows:

- **InvoiceNo**: Unique transaction ID; starts with **C** if cancelled.
- **StockCode**: Unique product/item code.
- **Description**: Product name/label.
- **Quantity**: Units purchased (negative = returns).
- **InvoiceDate**: Timestamp of the transaction.
- **UnitPrice**: Price per unit in GBP (£).
- **CustomerID**: Unique customer identifier.
- **Country**: Customer's country of residence.

The Online Retail II UCI dataset lines up directly with what this study is trying to do. With over a million transactions across two years, there's enough data to spot real patterns.

Each column pulls its weight. CustomerID and InvoiceDate together make RFM analysis since recency, frequency, and monetary value are all derivable from those two fields alongside Quantity and UnitPrice. InvoiceNo flags cancellations with the "C" prefix, which is useful for measuring returns and operational inefficiencies by category. StockCode and Description allow for breaking performance down by product and Country levels, adding a geographic layer for seasonal and regional trends.

The two-year window is also a good fit for seasonal analysis, being long enough to catch repeat cycles and test whether patterns like the Pareto distribution hold across different periods.

In short, the dataset covers customers, products, time, and geography. That's exactly what's needed to tackle behavior segmentation, category performance, and sales trends in one analysis.

- **Key Questions** (EDA Focus): What are the specific questions you will explore in the data?

#### **Segment aim Questions Key questions:**

- How many clients do we have? Where are these clients from?
- What is the average annual spending of my customers? How is the spending distributed? Are there any outliers?
- What is the average difference in days between the last and second last invoice notes of my customers? How is the difference in days distributed?
- How many times does each customer buy products from the store annually? What is the average order frequency?

#### **Operational efficiency key questions:**

- Are return rates higher in specific regions?
- What are the top ten most frequently bought products? Do they differ by region?
- Which countries generate the highest revenue vs. the highest order volume?

#### **Sales trends Key questions:**

- Are there seasonal trends observable across the two-year transaction period?
- Which products show strong seasonal demand versus consistent year-round demand?
- How does total revenue vary by month, quarter, and year?
- Does customer purchasing frequency change during peak periods?

- **Expected Insights:** What you expect to uncover

#### **Customer Segmentation:**

The RFM analysis is expected to reveal distinct customer segments based on how much they spend, how often they buy, and how recently they've been active. Some customers may score high on frequency but low on ticket size, buying often but in small amounts. Others may show the opposite, placing large orders occasionally. A smaller group is expected to stand out across all three dimensions: high spending, high frequency, and recent activity. These are the star

customers. Understanding what drives each segment, whether it's ticket size, order frequency, or consistent purchasing habits, is the core of what this analysis aims to uncover.

**Operational efficiency:**

Return rates likely cluster in specific regions or product categories rather than being evenly distributed. If certain countries generate disproportionate cancellations, that points to a logistics, quality, or expectation problem in that market. Top products probably follow a Pareto pattern, meaning that a small number of products drive the majority of volume, and that list will almost certainly shift by region.

**Sales trends:**

Revenue will most likely spike in Q4, consistent with the typical holiday season. Some products will show sharp seasonal peaks while others hold steady year-round. Monthly and quarterly breakdowns will reveal whether growth is genuine or just seasonal noise. Customer purchasing frequency during peak periods will show whether spikes are driven by existing customers buying more, or by one-time buyers who don't return.

- **Business Value:** What decisions could your insights support

**Customer segmentation:**

Knowing who the top customers are and how they behave allows the business to target the right segments, allocate resources where they matter most, and position its value proposition more effectively to maximize results.

**Operational efficiency:**

These findings tell the business where operational resources are being consumed without return, which products actually matter, and which markets are worth investing in versus pulling back from.

**Sales trends:**

Seasonal patterns justify targeted campaign timing, inventory pre-loading, and staffing decisions. If peak revenue depends on non-returning customers, retention becomes a priority. If loyal customers drive peaks, the focus shifts to deepening that relationship.